

Research paper

Genotypic and phenotypic diversity of *Mycobacterium tuberculosis* strains from eastern India

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ABSTRACT

Whole genome sequencing has been used to investigate the genomic diversity of *M. tuberculosis* in the northern and southern states of India, but information about the eastern part of the country is still limited. Through a sequencing-based strategy, this study seeks to comprehend the diversity and drug resistance pattern in the eastern region. A total of 102 *M. tuberculosis* isolates from North East ($n = 54$), and Odisha ($n = 48$) were sequenced along with 7 follow up isolates from Sikkim. The pre-XDR and XDR isolates diagnosed as per the NTEP diagnostic algorithm were subjected for phenotypic second-line liquid culture drug susceptibility testing in MGIT-960 system. After filtering out low quality isolates based on taxonomic classification and depth of coverage, variant calling was performed. We observed a high prevalence of multi-drug resistant TB (MDR-TB) lineage 2 (52/54) isolates in northeast whereas there was a mixed representation of lineage 1 (30/48) & lineage 3 (11/48) in Odisha. The MDR-TB isolates from Sikkim posed a high rate (51/53) of fluoroquinolone resistance and pairwise SNV distances (≤ 10) indicating possible local transmission events in the region. We observed occurrence of genetic variations in genes associated with bedaquiline and delamanid resistance. Our findings show the diversity of *M. tuberculosis* vary across the eastern regions, in north eastern states lineage 2 has a dominant presence while lineage 1 and 3 has mixed representation in Odisha. The high prevalence of fluoroquinolone resistance in north eastern region associated with variations in *gyrA* gene and may have been caused by local transmission events based on genomic similarities.

1. Introduction

According to the WHO tuberculosis report India is one of high tuberculosis (TB) burden countries with 2.8 million active cases (Global tuberculosis report, 2024). India is also listed among the 30 high multidrug-resistant TB (MDR-TB) countries with estimated 110,000 (Global tuberculosis report, 2024; India TB Report 2024 National TB Elimination Programme, 2024). Despite global efforts, only one-third of the patients with rifampicin-resistant TB (RR-TB) or MDR-TB received treatment (World Health Organization, 2022). The treatment success

rate of MDR-TB in India remains at 39.5 % (Shanmugam et al., 2022). Although the incidence rate of TB cases in India has declined over the years, the elimination of the disease will require understanding genomic diversity, evolution and transmission patterns.

The genotypic diversity of *Mycobacterium tuberculosis* (*M. tuberculosis*) across India is very dynamic and insight into genome-wide mutations remains limited (Singh et al., 2007). Recent whole genome sequencing (WGS) studies have unravelled the influence of lineage-specific mutations contributing to drug resistance phenotypes and associated with virulence (Shanmugam et al., 2022; Holt et al.,

Abbreviations: TB, Tuberculosis; *M. tuberculosis*, *Mycobacterium tuberculosis*; MTBC, *Mycobacterium tuberculosis* complex; SNV, Single nucleotide variant; CBNAAT, Cartridge-based nucleic acid amplification testing; PCA, Principal component analysis; ICMR, Indian Council of Medical Research; NTEP, National Tuberculosis Elimination Program; RR-TB, Rifampicin resistant tuberculosis; MDR-TB, Multidrug resistant tuberculosis; XDR-TB, Extensively drug-resistant tuberculosis; NE, North east India; MGIT, Mycobacteria Growth Indicator Tube; WHO, World Health Organization.

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2018). In recent times WGS has provided us with a better understanding of drug resistance phenotypes and genomic diversity of *M. tuberculosis* globally (Coll et al., 2018; Comas et al., 2013). Although studies representing the *M. tuberculosis* diversity in North and South India have been undertaken in recent years, genomic profiles representing Eastern and Northeastern (NE) regions yet to be explored (Advani et al., 2019; Chatterjee et al., 2017).

In our study, through a sequencing-based strategy we tried to comprehend the genomic diversity and drug resistance pattern of *M. tuberculosis* in the eastern region of India. We determined the predominant lineages followed by genetic diversity in these regions. We also compared the drug resistance by phenotypic DST and genotypic methods.

2. Methods

2.1. Sample collection and culture inoculation

Sputum samples from northeast and Odisha were sent to National Reference Laboratory, Regional Medical Research Centre, Odisha for confirmation of *M. tuberculosis* and drug resistance profile determination. All the sputum samples were initially treated with *N*-acetyl-L-cysteine-sodium hydroxide (NALC-NaOH) for decontamination and inoculated on Lowenstein-Jensen (LJ) slants for primary culture and mycobacteria growth indicator tube (MGIT) subsequently. Once the MGIT tubes showed detectable growth levels of the bacteria they were kept in an incubator at 37 °C for additional 2–3 weeks to achieve sufficient numbers of bacteria for genomic DNA isolation. A total 102 culture positive isolates were selected (convenient cohort sampling) from a pool of Cartridge-based nucleic acid amplification testing (CBNAAT) tested known Rifampicin resistance samples from North East regions and the Odisha isolates were suspected for tuberculosis infection. The isolates were collected between February 2017 to June 2020 and an additional 7 isolates were included in the study as follow-up collections of 5 cases.

2.2. GeneXpert testing

The sputum samples were first checked for the presence of *M. tuberculosis* using microscopy and then proceeded for drug resistance testing (Guideline for PMDT in India 2017 with messages, 2024). The samples were initially examined using the GeneXpert MTB/RIF CBNAAT assay for detection of TB and rifampicin resistance.

2.3. Line probe assay

If rifampicin resistance wasn't detected by GeneXpert, the sample was sent for first-line line probe assay (LPA) testing (Genotype *M. tuberculosis* DRplus version 2.0, Hain Life science GmbH) to determine isoniazid resistance status. Samples with detected resistance for first-line drug/s by GeneXpert and LPA were then tested by second-line LPA (GenoType *M. tuberculosis* DRsl by Hain Lifescience GmbH) for the detection resistance status of second-line drugs (fluoroquinolone and second-line injectables).

2.4. Culture based DST

MDR-TB and rifampicin resistance isolates diagnosed as resistant to second-line drugs (fluoroquinolone and second-line injectables) were further tested for phenotypic second-line drug susceptibility testing (DST) using 1 % proportion DST method by liquid culture in the Bactec MGIT 960 automated DST (MGIT-DST) system. Drugs including Kanamycin (2.5 µg/ml), Amikacin (1.0 µg/ml), Capreomycin (2.5 µg/ml), Ofloxacin (2.0 µg/ml), Levofloxacin (1.5 µg/ml), and Moxifloxacin (0.5 and 2.0 µg/ml) were used to test isolates. The phenotypic DST was carried out in accordance with the Global Laboratory Initiative (GLI)

Mycobacteriology laboratory manual and the National Tuberculosis Elimination Program (NTEP), India, training manual (Guideline for PMDT in India 2017 with messages, 2024; Global Laboratory Initiative Advancing TB Diagnosis, 2024; Training manual *M. tuberculosis* C DST:: Ministry of Health and Family Welfare, 2024).

2.5. Definition of drug resistance profiles

MDR: Isolate resistant to both rifampicin and isoniazid, **Pre-XDR:** Isolate resistant to isoniazid and rifampicin and resistant to any fluoroquinolone, **XDR:** Isolate resistant to any fluoroquinolone (levofloxacin and/or moxifloxacin) and to at least one of three second-line injectable drugs (SLIs: capreomycin, kanamycin and amikacin), in addition to multidrug resistance (Meeting Report of the WHO Expert Consultation on the Definition of Extensively Drug-Resistant Tuberculosis, 2021).

2.6. DNA extraction and sequencing library preparation

The DNA from all the isolates were extracted using bead beating method using HiPurA bacterial genomic DNA purification kit in the Biosafety level 3 (BSL-3) facility of ICMR-RMRC, Bhubaneswar. The DNA quality of individual tuberculosis samples were checked using NanoDrop™ spectrophotometers and samples having a 260/280 ratio between 1.6 and 2.0 were selected for preparation of whole genome sequencing libraries. All sequencing libraries were prepared using Nextera XT DNA library preparation kit from Illumina, Inc. with 1 ng of input DNA in 0.2 ng/µl concentration. The prepared whole-genome sequencing libraries were pooled and sequenced using NextSeq 550 system with a 2 × 150 bp paired-end layout.

2.7. FASTQ quality control and taxonomic classification

The raw sequences were first checked with FASTQC followed by quantification of *M. tuberculosis* complex (MTBC) using Kraken2 against a database containing archaea, bacteria, viral, plasmid, vector and human reference assembly (Wood et al., 2019; Babraham Bioinformatics - FastQC A Quality Control tool for High Throughput Sequence Data, 2024). Isolates having less than 80 % of reads classified as MTBC were removed from further analysis.

2.8. Whole genome sequence analysis

The sequenced and curated whole genome sequences were mapped to H37Rv reference genome (NC_000962.3) using the BWA-MEM algorithm (Li, 2013). Isolates having less than 90 % of the genome covered with at least 10× depth of coverage were removed from further analysis (Supplementary Fig. 1B). Picard was used to remove duplicate reads before variant calling (Walker et al., 2014a). Single nucleotide variants (SNV) and short insertion and deletions were called using Pilon with -fix all, breaks option. The SNV and INDELs were extracted from the raw VCF file using Bcftools. A minimum map quality (MQ) score of 20, minimum base quality (BQ) score of 20, minimum read depth (DP) of 5, and fraction of reads supporting allele ≥ 0.80 used to select the fixed variants. The filtered variants from all the isolates were merged using Bcftools and annotated using SnpEff (Danecek et al., 2021; Cingolani et al., 2012).

2.9. Lineage calling

Lineage determination of all the isolates were performed using fast-lineage-caller from Farhat-lab (Freschi et al., 2021). The tool takes VCF files as input and assigns lineages based on presence of pre-defined lineage determining mutations reported by multiple sources. We also predicted digital spoligotyping using SpoTyping tool and the predicted types were annotated by querying SITVIT database, both of the methods supported the lineage calls, and Fast lineage caller with Coll et al. set

were considered in the final analysis (Freschi et al., 2021; Xia et al., 2016; Coll et al., 2015). Isolates with multiple lineages assigned were treated as mixed infection and removed from further analysis.

2.10. Curation of North and South Indian samples

The north Indian whole genome sequences ($n = 200$) from Advani et al. were downloaded from NCBI SRA database and processed using the same workflow as sequenced samples described earlier. Variant call files of the South Indian samples ($n = 223$) from Manson et al. were available from Broad Institute portal (https://olive.broadinstitute.org/collections/tb_india.1) but with a reference genome. The variant positions were updated to NCBI H37Rv reference assembly using Picard LiftOver utility and CP003248.2 to NC_000962.3 conversion chain file generated with UCSC toolkit. Finally 365 samples out of 423 were retained after sample quality filtration.

2.11. Phylogeny reconstruction

Phylogenetic analyses were performed using only the SNV sites after removing variants detected the PE, PPE, and PGRS regions using Bedtools (Quinlan and Hall, 2010). Merged multi-sample VCF files were converted to fasta format containing the SNVs using vcf2fasta tool. The

construction of phylogenetic tree was performed using IQ-TREE2 (Minh et al., 2020). We have restricted model selection to GTR models and the best model was selected with an implementation of ModelFinder Plus. SH-aLRT test and 1000 bootstrap replicates were computed using UFBoot2 implementation in the tool (Kalyanamoorthy et al., 2017; Hoang et al., 2018).

2.12. Genotype principal component analysis

The genotype principal component analysis (PCA) was computed with all isolates from India combined with 2500 isolates collected from GMTV database (Chernyaeva et al., 2014). Pairwise distances among the isolates were calculated using snp-dist tool. Principal components were calculated and plotted using PCATools R package.

2.13. Statistical analysis

Except for the mentioned cases, the rest of the statistical tests were performed using base R functions and plotted using the GGPUBR wrapper of the ggplot2 package.

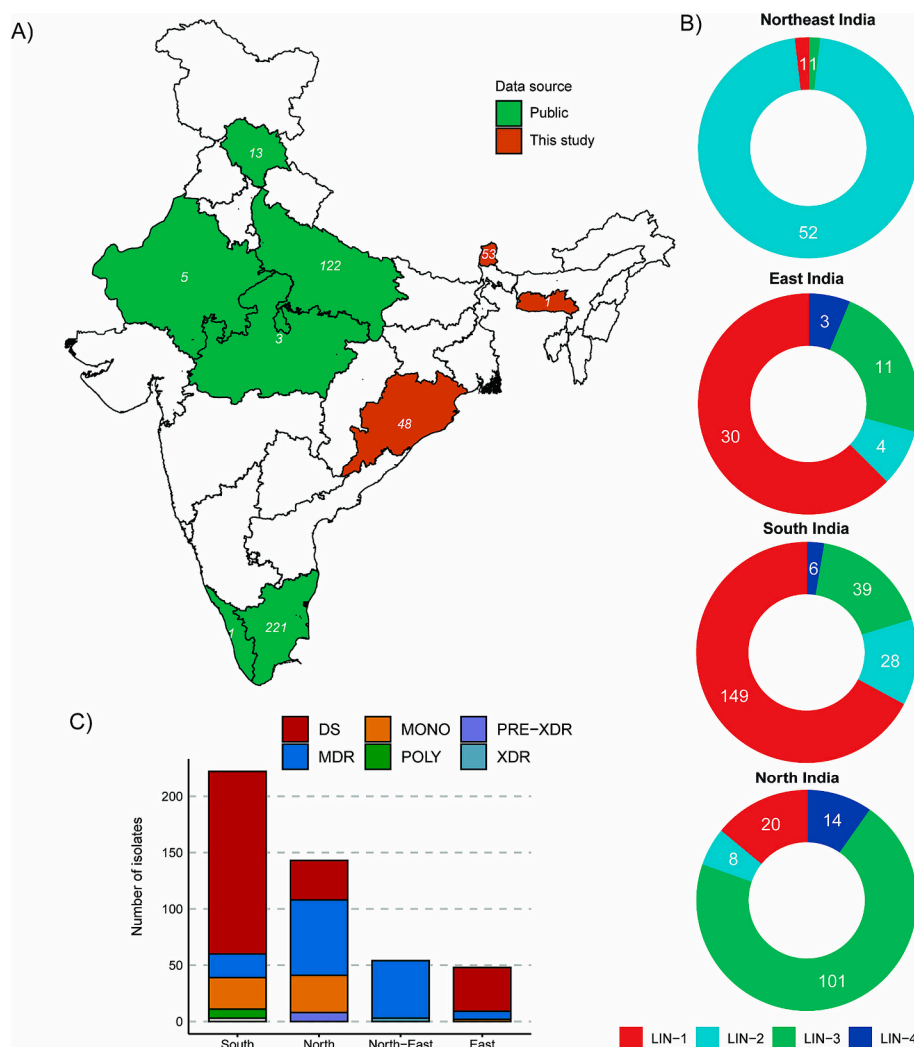


Fig. 1. A) India map depicting the state-wise location of *M. tuberculosis* whole genome sequence isolates from our study and the isolates collected from other published studies for comparative analysis, B) Plots showing the distribution of *M. tuberculosis* lineages across different regions with number of isolates, and C) The drug resistance profiles of sequenced isolates and the representative sequences across India.

3. Results

A total of 118 isolates were sequenced out of which 111 were unique isolates and 7 follow-up isolates corresponding to 5 cases. Out of 111 samples 102 samples passed sequence quality control criteria and used for the primary analysis. The suspected MDR-TB phenotype for 64 cases were initially confirmed for RR-TB using the CBNAAT test and later tested for isoniazid and rifampicin resistance using first-line LPA. A total of 50 cases from NE showed the presence of fluoroquinolone resistance based on second-line line probe assay (SL-LPA) and culture-based DST results. In addition to sequenced samples we curated 365 whole genome sequences representing diversity in South and North India and compared with the newly sequenced samples (Fig. 1A). The median age of the patients was 35 years and the number of male and female individuals was 57 and 44 respectively (Table 1, Supplementary Fig. 1A).

3.1. *M. tuberculosis* lineage diversity in the NE and eastern region

We observed a contrasting difference between the lineage distribution of the NE region (Sikkim and Meghalaya) and Odisha. Lineage 2, commonly known as Beijing lineage has a very dominant presence in NE (52/54) whereas lineage 1 (Indo Oceanic lineage) is more prominent 30/48 in Odisha (Table 1, Fig. 1B). Although, lineage 1 is very dominant in Odisha, the region also has a presence of lineage 3 (11/48), also known as Central-Asian lineage followed by lineage 2 and lineage 4 (European/American) strains (Fig. 1B, Supplementary Table 1). The lineage diversity of Odisha is very similar to the diversity of South India in contrast to North India (Fig. 1B). The isolates from the southern part of Odisha belonged to lineage 1 whereas central and eastern districts show a mixed presence of all three lineages. Based on CBNAAT & LPA assays the isolates from the NE were mostly classified as MDR-TB and strains from Odisha were in the drug-sensitive category, the curated sequences representing South India were drug-sensitive whereas the North India MDR-TB was highly represented (Fig. 1C).

3.2. Phylogenetic classification and genetic similarity with other regional and global strains

To better understand the genomic diversity in the eastern regions compared to the rest of the country, we constructed a phylogenetic tree with SNVs (Fig. 2). We found that the drug resistant *M. tuberculosis* isolates from Sikkim were divided into two subgroups while coinciding

Table 1

Demographics of patients enrolled for the study, drug resistance profiles (genotypic) of sequenced isolates (excluding follow-ups) and lineage distribution (detected by variant calls against published panels).

	Count	Percentage (%)
Age (yr)		
<30	41	40.20
>30	61	59.80
Sex		
M	57	55.88
F	44	43.14
Undisclosed	1	0.98
Region		
Northeast	54	52.94
East	48	47.06
Resistance (WGS)		
DS	39	38.24
MONO	2	1.96
MDR	58	56.86
XDR	3	2.94
Lineage		
Lineage 1	31	30.39
Lineage 2	56	54.90
Lineage 3	12	11.76
Lineage 4	3	2.94

with other lineages 2 isolates from Tamil Nadu and Uttar Pradesh. The isolates from Odisha were distributed between lineage 1 and lineage 3 clusters and supported our regional influence hypothesis (Fig. 2). The lineage 4 isolates from Odisha clustered with strains similar to Uttar Pradesh and Rajasthan (Fig. 2). Furthermore, to extend the understanding of the genetic diversity of the *M. tuberculosis* strains in the global context, we performed PCA of pairwise SNV distances combining isolates from India with a set of 2215 global *M. tuberculosis* curated from the GMTV database (Chernyaeva et al., 2014). Principal components 1 and 2 explained more than 96 % of the variance in the pairwise SNV dataset (Fig. 3A). The *M. tuberculosis* isolates from India grouped with the respective global lineages (Fig. 3A). We selected a representative subset ($n = 153$) of global strains from Comas et al. study to access the similarity with the strains from eastern India (Comas et al., 2013). The lineage 1, 3, 4 strains showed a very high SNP differences (mean > 300) in contrast to lineage 2 strains respective to their global representations (Supplementary Fig. 2). The SNP differences indicate that lineage 2 isolates from eastern India is very similar to their closest global relatives from south-east Asia.

It is hypothesized that terminal branch lengths are inversely proportional to the spreading capability of a pathogen (Freschi et al., 2021). When we quantified all the branch lengths of terminal nodes of eastern India strains, we observed that the terminal branch lengths of lineage 2 was significantly smaller than lineage 1, 3, and 4. The median branch lengths of lineage 2 strains were 8.06×10^{-06} whereas the median branch lengths of lineage 1, 3, and 4 were 2.1×10^{-04} , 1.36×10^{-04} , and 1.8×10^{-04} respectively (Fig. 3B).

3.3. Local transmission of *M. tuberculosis* in Sikkim

Based on the phylogenetic clustering we observed the presence of closely related lineage 2 sequences from Sikkim. To understand the degree of relatedness among the strains we calculated pairwise SNP differences of all 102 whole genome sequences and clustered them (Fig. 3C). Based on the clustering pattern we observed two distinct groups of lineages 2 isolates with very less SNP differences (0–428, median 32) among them compared to other strains (Supplementary Fig. 3). Using ≤ 12 SNP differences, a threshold previously used to define recent transmission cluster (Walker et al., 2014b), 29 isolates from Sikkim were identified with probable epidemiological links. To further resolve the genetic similarity among the strains we created a phylogenetic tree of all 53 isolates from the region and included their district-level information. The resultant phylogenetic tree represented the main cluster ($n = 42$) consisting of strains mostly from east Sikkim with 3/42 strains each from north and west Sikkim. The smaller cluster ($n = 9$) represented sequences from the east district along with 3/9 strains from the west district (Fig. 3D).

3.4. Genomic basis of high fluoroquinolone resistance in NE India

The first-line resistance profiles isolates were determined using genotype drug susceptibility testing (gDST) and culture base DST was done for 6 s-line drugs consisting of fluoroquinolones and second-line injectables. The CBNAAT results completely overlapped with first-line LPA results and 61/102 isolates were rifampicin resistant followed by 60/102 isolates with isoniazid resistance. Second-line LPA results showed 54/102 isolates with fluoroquinolone resistance and 6 samples having injectable drug resistance. Among the fluoroquinolone resistant isolates, a majority 28/54 of them showed resistance to levofloxacin in culture-based DST test. In the injectables category all the 6 samples showed resistance to kanamycin.

The concordance between prior phenotyping and mutational profile based phenotypes were checked using two *M. tuberculosis* mutational catalogue i.e. WHO TB Mutation catalogue and TB-profiler. Based on the resistance-conferring variant statistics, strains from Sikkim and Meghalaya had higher numbers of associated mutations compared to

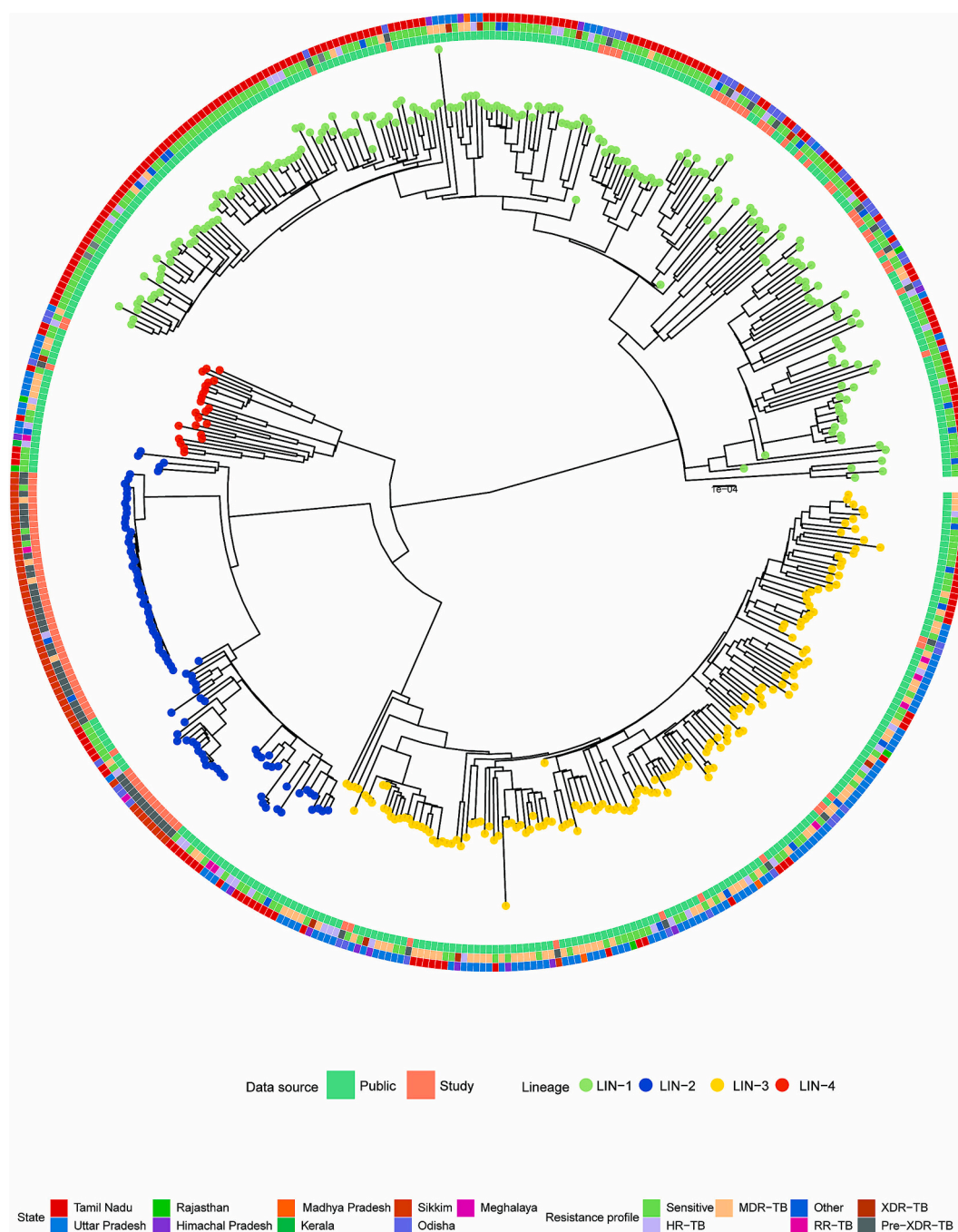


Fig. 2. Neighbour joining tree representing *M. tuberculosis* isolates from North, South, Northeast and East India ($n = 467$), with 1000 bootstraps (outgroup *M. canetti* not shown). The node colour represents isolate lineage, inner circle shows the source of the WGS sample followed by corresponding drug resistance phenotype patterns and outer circle coloured based on the sample collection location (divisions/states).

Odisha (Fig. 4A). We found that out of 54 isolates collected from NE, 51 were mostly classified as MDR-TB and 3 as extensively drug-resistant XDR based on primary resistance profiling. The isolates collected from Odisha were drug-sensitive, out of the 48 isolates, 7 were classified as MDR-TB and 2 as mono-resistant. Based on first-line LPA assay results, the initial understanding of antibiotic resistance phenotype was limited to isoniazid and rifampicin but phenotypes predicted using WGS showed additional resistance to pyrazinamide, ethambutol, and streptomycin in MDR-TB isolates (Fig. 4B, Supplementary Fig. 4 A-D). Rifampicin and isoniazid resistance was mostly explained by *rpoB*.p.Ser450Leu and *katG*.p.Ser315Thr respectively. We also observed the presence of drug resistance-associated singleton variation in *kasA*, *rpoB*, *katG*, and *ahpC*

gene in 5 of the isolates. In terms of second-line resistance, the MDR-classified isolates harboured mutations in the *gyrA* and *gyrB* genes. Three missense mutations *gyrA*.p.Ala90Val, *gyrA*.p.Asp94Gly and *gyrA*.p.Asp94Tyr conferred the majority of the resistant phenotypes (Fig. 4C). Using the sequencing-based drug resistance information, 55 out of 59 MDR-TB isolates were re-classified as pre-XDR-TB as they are resistant to all first-line drugs and fluoroquinolones accordance to the new WHO guidelines (Fig. 4D). Out of the 7 follow-up isolates one case acquired cycloserine resistance conferred by *alr*.p.Leu113Arg.

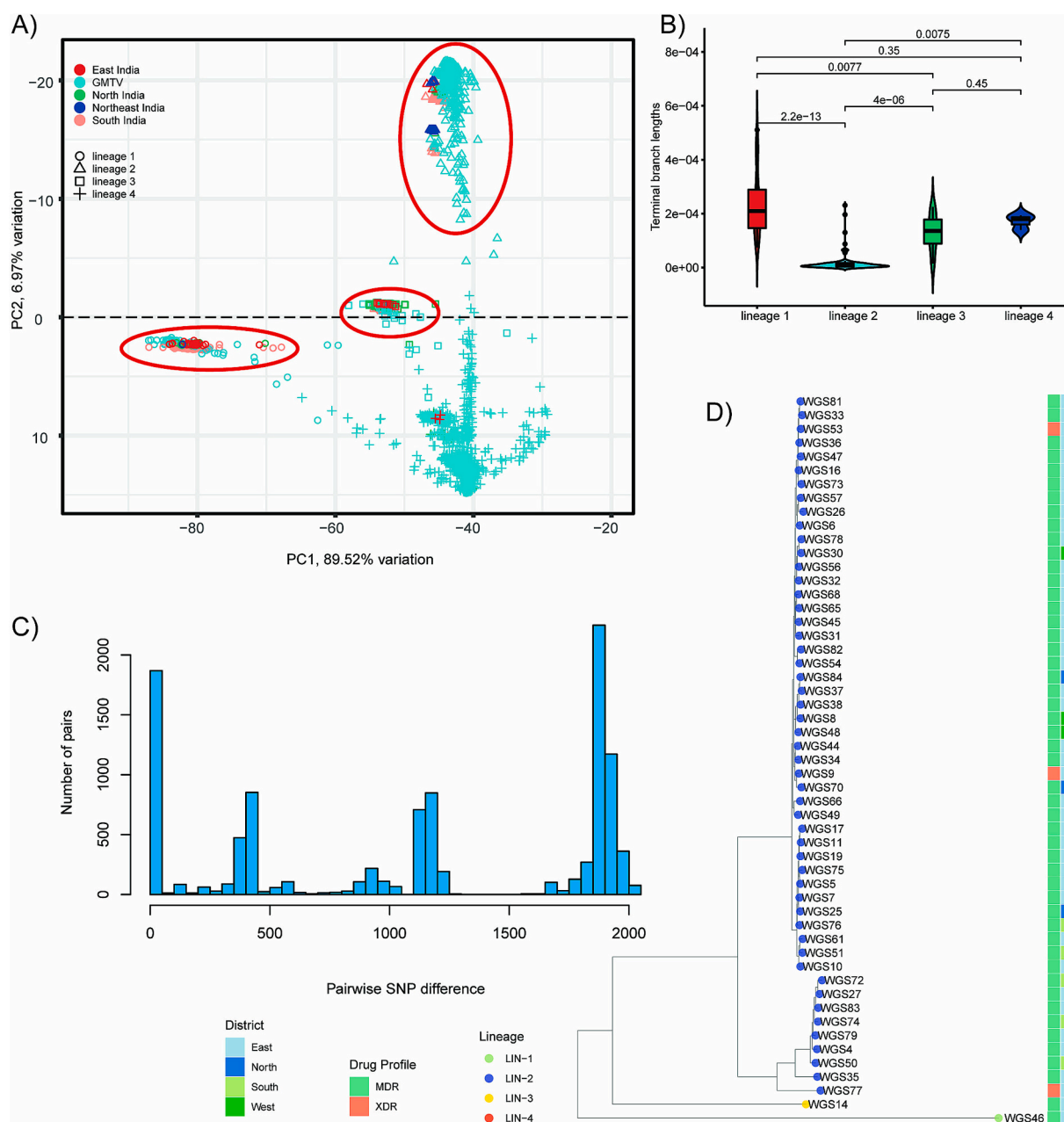


Fig. 3. A) Principal component analysis of Indian TB strains with GMTV strains. The shape of the dots represent the lineage and colour represents the dataset/strains source, B) Terminal branch length distribution of LIN-1:4 with pairwise Wilcoxon rank sum test performed to test the difference between distributions, C) Histogram of pairwise SNP differences among the 102 samples, X-axis represents SNP differences and Y-axis represents the number of pairs. D) Phylogenetic tree representing the Sikkim sample local transmission clusters with their drug resistance phenotypes, lineage and the sample collection district.

3.5. Mutations in other resistance-associated genes

By comparing the annotated nucleotide alterations with the most recent version of Colls set (Coll et al., 2015) of mutations related to drug resistance, we identified MDR-TB cases harbouring resistance to para-aminosalicylic acid and linezolid. Though the association of *atpE*, *mmpR* (*Rv0678*), and *mmpL5* mutations with bedaquiline resistance is yet to be established with a larger set of evidence, it is assumed that any variation leading to loss of function (LoF) mutation may lead to resistance development. When we screened our isolates for any changes in the *Rv0678* gene, we found a total of 45 isolates harbouring at least one missense variant. Three lineages 1 isolate from Odisha have *mmpR* p. Gly162Glu, out of which 2 were determined drug-susceptible and 1 had MDR-TB phenotype. The remaining 42 isolates from Sikkim belonging to

lineage 2 are depicted to have *mmpR* p. Glu55Asp change and were mostly MDR-TB cases (35/42). The remaining 7 isolates have mono-resistance to isoniazid or rifampicin, 3 were drug-sensitive, and 2 were resistant to additional second-line drugs. Accompanying the *mmpR* gene we found all 102 isolates harbour at least one *mmpL5* missense variant and novel *mmpL5* p. Arg363Trp variation.

While looking into delamanid-associated gene mutations we came across three novel missense variations *fbiA* p. Arg165Glu, *fbiB* p. Thr302Ile, and *fbiC* p. His772Tyr. The *ddn* gene only contained two synonymous nucleotide changes. In addition to the WHO mutation catalogue we also searched for these variations in Indian catalogue of *M. tuberculosis* mutations and didn't observe any of the mutations being associated with resistance phenotypes (Indian catalogue of *Mycobacterium tuberculosis* mutations and their association with drug resistance,

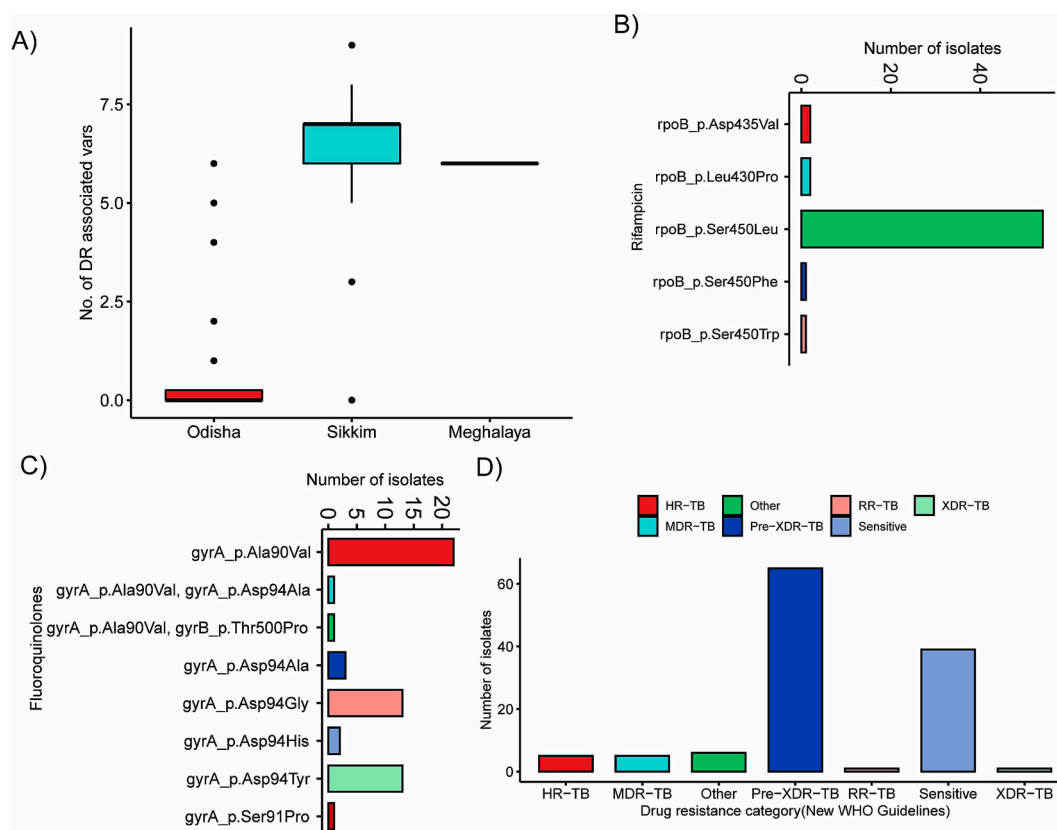


Fig. 4. A) Distribution of detected number of drug resistance associated variations according to the respective region, B–C) Drug resistance associated genes and amino acid changes with respective mutation frequency in sequenced isolates.

2022).

4. Discussion

The primary focus of our study was to extend the understanding of *M. tuberculosis* genomic diversity in the eastern parts of India. For the first time, our study shed light into the lineage diversity of *M. tuberculosis* in Odisha. Additionally, our study shares insight about the molecular basis of high prevalence of MDR-TB fluoroquinolones-resistant TB in the NE region.

We observed probable occurrence of localised transmission clusters of MDR-TB lineage 2 strains (≤ 12 SNPs) in NE region. Contact-based transmission of MDR-TB strains might be one of the reasons for the high abundance of fluoroquinolone resistance along with rifampicin-resistant cases in the region. The cluster in East Sikkim indicated the probable presence of local transmission clusters and require to be taken into account for prevention programs (Pai et al., 2006). We observed that the terminal branch lengths of lineage 2 strains were significantly smaller than other lineages indicating higher transmissibility and can be the probable reason we observe less SNP differences in Sikkim cluster (Freschi et al., 2021). Although, based on genomic similarity we observed probable local transmission events but due to lack of respective epidemiological information it cannot be concluded with certainty. The divergence from global strains was more in lineage 1, 3 and 4 isolates while lineage 2 isolates shared genetic similarity with their south east Asian relatives.

The prevalence of Beijing lineage in NE regions is coherent with a recent spoligotyping survey conducted in Sikkim (Devi et al., 2021). Devi et al reported a 62.41 % occurrence of lineage 2 strains in Sikkim based on their sampling.

A recent study examining the molecular diversity of *M. tuberculosis* in the NE region revealed a high prevalence of lineage 2 (62 %) with 29.7

% MDR-TB phenotype in contrast to other *M. tuberculosis* lineages (4.7 %), the overall resistance rate to be 20.3 % (Devi et al., 2021). The NE region is known to have a very high burden of drug-resistant tuberculosis cases (Devi et al., 2021; Mudliar et al., 2022). Although our Sikkim-based isolates were enriched for the rifampicin (CBNAAT positive) resistance genotype, contemporaneous research revealed that our results are indicative of the NE region and point to the predominance of the Beijing lineage having a multi-drug resistance phenotype (Devi et al., 2021). The district-wise prevalence of lineages suggests the genetic diversity of *M. tuberculosis* is influenced by diversity of the neighbouring regions.

On the other hand, in Odisha, we observe a very diverse representation of *M. tuberculosis* strains, all three lineages prevalent across India are observed in this region. Lineage 1 is the dominant one followed by lineage 3 and lineage 2. The distribution of lineage has a similar pattern as the study conducted in Tamil Nadu, the cause might be the frequent migration of people towards southern regions in search of livelihood. Although Odisha has a less burden of drug-resistant tuberculosis, the overall number of reported cases is also less due to lack of healthcare facilities in close vicinity, and lower per capita income (Hussain et al., 2020). Odisha is a tribal population-dominated region and the community is scattered across dense forest areas or mountains making it very difficult to estimate the actual burden of tuberculosis. Studies performed in the nearby state of Madhya Pradesh stated a high tuberculosis burden among the Saharia tribe (20.4 %) a geographically isolated population of agriculture laborers (Rao et al., 2019).

Detection of drug-resistant phenotype in a very early stage of diagnosis helps in the determination of the treatment regimen, which is a major contributor to the treatment outcome. Due to the huge burden of new cases, sometimes confirmatory tests for second-line drug resistance take up to 20–30 days after the initial detection of rifampicin resistance using rapid PCR-based techniques. Recent studies have demonstrated

the reduced efficacy of GeneXpert *M. tuberculosis*/RIF, Hain *M. tuberculosis* DRplus, and Hain *M. tuberculosis* DRsl in the detection of drug antibiotic resistance due to region/lineage-specific mutations that are not incorporated in the tests (Manson et al., 2017).

In the sequenced isolates, we were able to explain of the drug resistance phenotypes with the SNVs and even extended the antibiograms of MDR-TB isolates. A large proportion (48/54) of rifampicin and fluoroquinolone-resistant (MDR) isolates harboured *gyrA* E21Q, D94A, G668D, and A90V along with *rpoB* S450W, *rpoC* G332S amino acid substitutions. Additionally, while looking for variations in the bedaquiline and delamanid associated genes we found two non-synonymous changes in a total of 45 MDR-TB cases. The *Rv0678* p.Gly162Glu change was already reported as a novel mutation in the ICMR-JALMA cohort however we found *Rv0678* p.Glu55Asp change in the same gene is not reported in any other study from India (Advani et al., 2019). Despite of a greater representation (42/62) of the variation in lineage 2 isolates an association was not established due to the unavailability of phenotypic resistance profiles.

5. Conclusions

In conclusion our study extended the understanding of *M. tuberculosis* lineages distributed across eastern region of India. This is one of the first study to discuss genetic diversity in the eastern state of Odisha. The prevalence of high fluoroquinolone resistance among lineage 2 strains from Sikkim is associated with local transmission events and high burden of *gyrA* mutations. In addition to the pre-XDR phenotype associated variations, single nucleotide variants were detected in bedaquiline resistance associated gene. Further extensive studies needed to be undertaken to validate and better understand the transmission patterns and antimicrobial resistance phenotypes evolving in the regions.

5.1. Limitations

The representation of *M. tuberculosis* diversity in eastern part of India is limited to a small number of isolates from Sikkim and Odisha. A better understanding of the regional diversity will require representative sequences from all the states. The phenotypic drug resistance profiles of the *Mycobacterium tuberculosis* bacteria was limited and the resistance profiler was later mainly determined using whole genome mutation profiles.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.meegid.2025.105713>.

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Code availability

The codes used for analysis are available at (https://github.com/arupghosh/tb_india_genomics).

Ethics approval

This study was approved by the human ethics committee of ICMR-Regional Medical Research Centre, Bhubaneswar (ICMR-RMRCB/IHEC-2019/039) and written consent forms are archived as a part of National Reference Laboratory record.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

CRedit authorship contribution statement

Arup Ghosh: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Viplov Kumar Biswas:** Writing – review & editing, Resources, Methodology, Investigation. **Himadri Bhusan Bal:** Validation, Resources, Project administration, Investigation. **Dasarathi Das:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Sanghmitra Pati:** Writing – review & editing, Supervision, Investigation. **Bhawna Gupta:** Supervision, Investigation. **Sunil Kumar Raghav:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors have declared no conflict of interests.

Data availability

The raw FASTQ files are deposited to NCBI SRA repository under accession id PRJNA896734.

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